

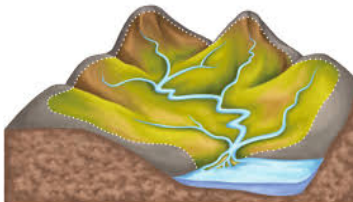
Watershed Segmentation Algorithm

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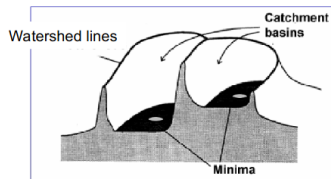
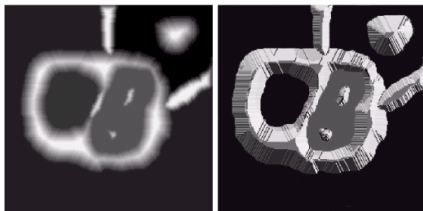
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- ▶ In watershed segmentation algorithm, the process of constructing dam(to stop merging two water bodies) is mimicked on images
- ▶ Consider a landscape, and visualize the topography of the landscape



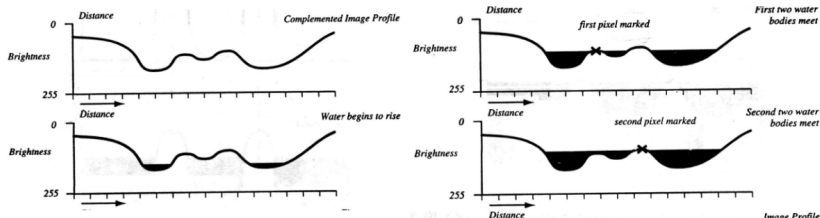
- ▶ In such a topographic view, there are 3 types of points when rain water drop falls:
 - Water drop falls at regional minima point P
 - Water drop falls at a point P such that the water drop flows from P to a single minimum. (The catchment point)

- Water fall at a point P such that the water drop would be equally likely to flow from P to more than one minimum. (The divide line point or watershed line point or dam point.)
- ▶ The set of all dam points is called as dam or watershed line or divide line



How to find watershed lines, given the topography?

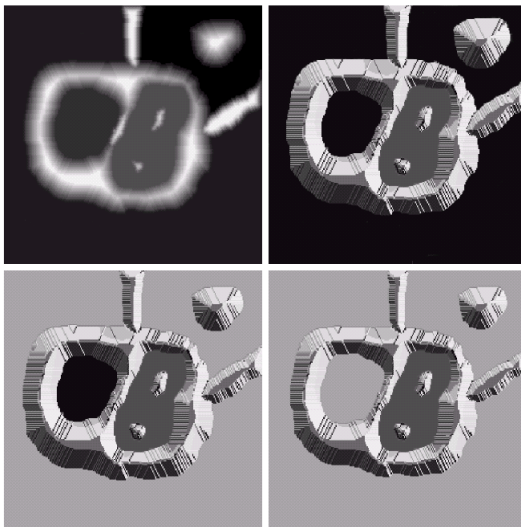
- ▶ The idea is simple:
 - Suppose that a hole is punched in each regional minimum and that the entire topography is flooded from below by letting water rise through the holes at a uniform rate.
 - When the rising water in distinct catchment basins is about to merge, a dam is built to prevent merging.
- ▶ 1D-profile of the topography along with water filling and dam construction are shown below



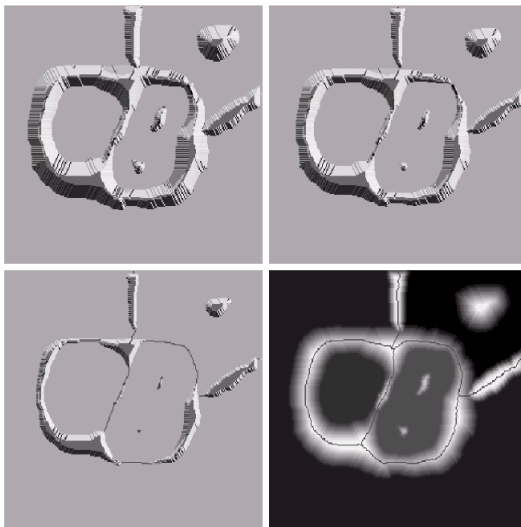
a b
c d

FIGURE 10.44

(a) Original image.
(b) Topographic view.
(c)–(d) Two stages of flooding.



Watershed segmentation (cont.)



e f
g h

FIGURE 10.44

(Continued)

(e) Result of further flooding.
(f) Beginning of merging of water from two catchment basins (a short dam was built between them). (g) Longer dams. (h) Final watershed (segmentation) lines. (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)



To mimic the process of building dams as explained above for segmenting images

- ▶ Visualize an image in 3D: pixel location(2D) and pixel value(third dimension)

Watershed Segmentation on Gradient Magnitude of Image G

- ▶ Find local minimas in G
- ▶ For each of the local minima, assign distinct labels
- ▶ For each intensity level $k=0$ to Max in g
 - For each unlabeled neighbour p (with value k) of each of the labeled pixels
 - ▶ If p is adjacent to pixel IN g with a single label I , then assign label I to p
 - ▶ Else(adjacent to at least two pixels with different labels), Do not label p , and call that p as a point in the watershed line or dam point.



- Output all pixels IN g , their corresponding labels, and also the set all of dam points



- ▶ Can we use watershed segmentation algorithm directly on the input gray scale image
 - No, It does not produce desired results
- ▶ Watershed algorithm can be used on the gradient image instead of the original image.
 - As per the underlying assumption each object corresponds to one catchment area
 - The gradient at the bottom of catchment area is zero

a b
c d

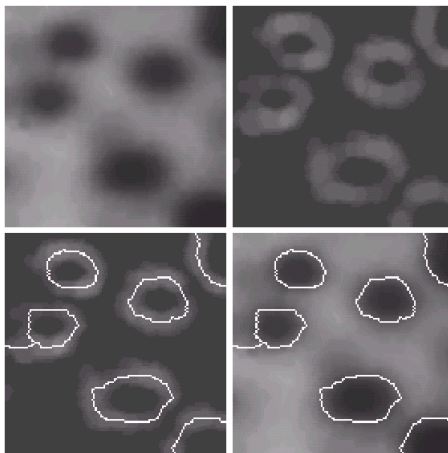
FIGURE 10.46

(a) Image of
blobs. (b) Image
gradient.

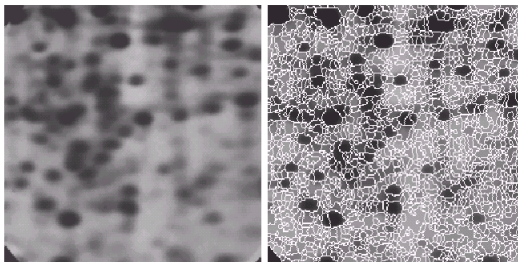
(c) Watershed
lines.

(d) Watershed
lines
superimposed on
original image.

(Courtesy of Dr.
S. Beucher,
CMM/Ecole des
Mines de Paris.)



Due to noise and other local irregularities of the gradient, oversegmentation might occur.



a b

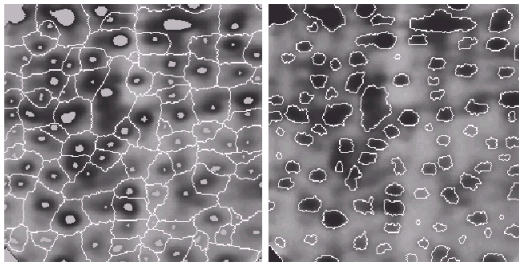
FIGURE 10.47

(a) Electrophoresis image. (b) Result of applying the watershed segmentation algorithm to the gradient image. Oversegmentation is evident.

(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)



- ▶ How to avoid over segmentation
 - Limit number of local minimas
 - One way to limit number of local minimas is to use **markers**
 - ▶ Marker is a connected component with similar(close) values in the gradient image
 - Consider all points in the same marker as the same local minima (they all need to be labelled with the same label)



a b

FIGURE 10.48

(a) Image showing internal markers (light gray regions) and external markers (watershed lines).
(b) Result of segmentation. Note the improvement over Fig. 10.47(b).
(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

Step 1: Convert to binary image such that the foreground is in white, and the background is in black



Figure 1: Input Image

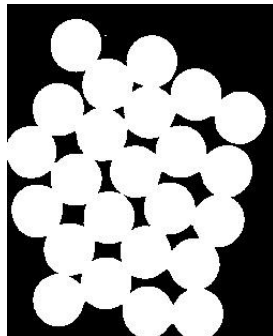


Figure 2: Binary Image with foreground in white



Step 2: Remove noise using opening operation

Step 3: Find sure-Foreground(Fg) and sure-Background(Bg) on output of step 2

- ▶ To find sure-background, do dilation(black pixels are sure Bg)
- ▶ To find sure-Fg
do erosion(white pixels are sure Fg) (works well when objects are not touching)
or
Thresholded distance transform output(works well when objects are touching), where distance transform: $\min d((x,y), Bg)$ for each pixel (x,y)

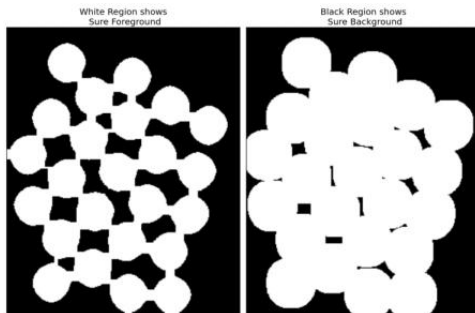


Figure 3: Left: Eroded; Right: Dilated



Observation after step3:

- ▶ There are three types of pixels
 - Sure Fg pixels
 - Sure Bg pixels
 - Non-labeled pixels
- ▶ The Non-labeled pixels will be classified to Fg or Bg by the watershed segmentation algorithm

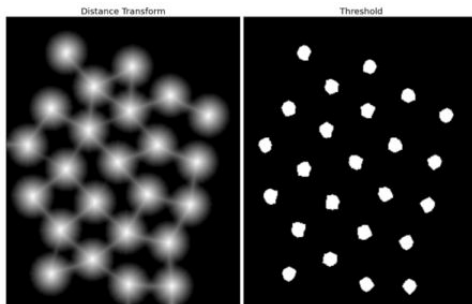


Figure 4: Left: Distance Transform; Right: Thresholded Distance Transform

Step 4: Find the connected components in the sure-Fg image

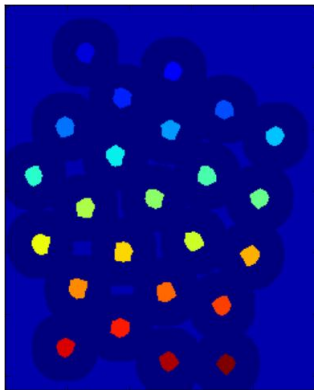


Figure 5: Each connected component is shown in different colour

Step 5: Apply watershed algorithm, keeping each marker as local minima

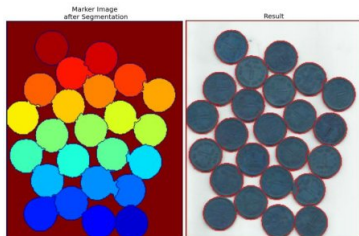


Figure 6: Each connected component is shown in different colour



1. Convert the given colour image to a **grayscale image**.
2. Convert the grayscale image to a **binary image** using thresholding:
 - Objects (foreground) $\rightarrow$ 1
 - Background $\rightarrow$ 0
3. Apply the **Distance Transform** to the binary image.
 - It computes the distance of each foreground pixel from the nearest background (zero) pixel.
4. Threshold the distance-transform image to obtain the **sure foreground**.
 - These regions represent the central parts of the objects.
5. Dilate the binary image to obtain the **sure background**.
 - Dilation expands the foreground regions and identifies regions known to belong to the background.
6. Obtain the **unknown region**:



- Subtract the sure foreground from the sure background.
- 7. Compute **connected components** on the sure-foreground image.
 - Assign a unique label to each connected component.
 - These labeled connected components are used as **markers**.
- 8. Compute the **Gradient Magnitude Image** of the grayscale image.
- 9. Apply the **watershed** algorithm on the Gradient Magnitude Image, considering markers in place of local minima
 - In basic watershed, flooding starts from the **local minima** of the topographic surface.
 - In marker-controlled watershed, the **markers act as selected flooding sources**, effectively replacing the unwanted local minima.
 - Flooding starts from these markers and expands into neighboring regions.
 - Watershed boundaries are formed where different flooded regions meet.